Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 1 Quiz 1 - Kinematics in One Dimension

Mock Exam A · Form 1: Foundations

Multiple Choice Questions: Select the best answer. (1 point each)

1. A student walks right and returns to the starting mark. Which quantity must vanish?

(A) maximum speed

(B) distance

(C) average speed

(D) average velocity

2. A cart initially moves at +4 m/s and accelerates constantly at +1 m/s² for 3 s. What is its change in velocity?

(A) -3 m/s

(B) 6 m/s

(C) 1.5 m/s

(D) 3 m/s

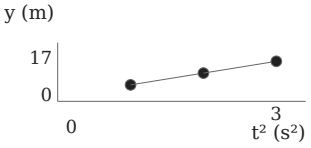
3. A sensor records downward position for a weight dropped from rest. Use g=10 m/s². t² (s²) -> y (m): (1, 7); (2, 12); (3, 17). Which vertical reading does the model predict at zero horizontal coordinate?

(A) 2

(B) 4.5

(C) 3.25

(D) 0.75



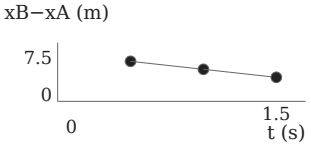
4. A runs at +5 m/s and B at +2 m/s. Record the lead of B ahead of A as a signed separation. t (s) -> xB-xA (m): (0.5, 7.5); (1, 6); (1.5, 4.5). What is the signed change in the vertical reading from first to last trial?

(A) -0.6

(B) -3

(C) -1.8

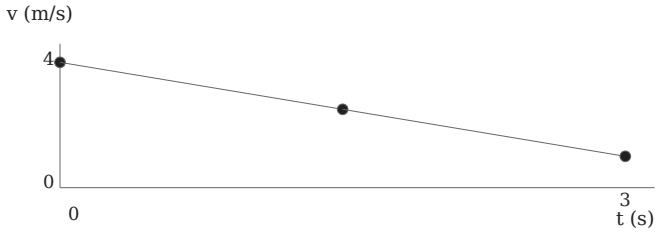
(D) -4.2



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A sensor records a cart with x0 = -2 m and v0 = +4 m/s. It has constant a = -1 m/s² for 3 s. Right is positive.

a. Describe the shapes and initial values of x–t, v–t, and a–t. **(1 pt)**



b. Find final position, displacement, and distance; explain any reversal. **(3 pt)**

6. Photogates record the position of a cart traveling at constant velocity. t (s) -> x (m): (0.5, 7); (1, 6); (1.5, 5).

a. Build the model for track photogates from physics and predict both slope and intercept. **(2 pt)**

b. Plot the first and last measurements and describe the trend. **(2 pt)**

7. A cart moves at constant velocity v for time t. Derive displacement from v–t area, distinguishing distance from displacement when v is negative. **(3 pt)**